

Automated Blood Processing Equipment Market Size, Share, Growth 2026-2035

The global automated blood processing equipment market is growing steadily due to rising demand for safe blood handling systems and advanced diagnostic technologies. The market was valued at USD 2.23 billion in 2025 and is projected to reach USD 4.68 billion by 2035, expanding at a CAGR of around 7.7% during the forecast period from 2026 to 2035.

Increasing blood donation activities, growing surgical procedures, and the adoption of automated healthcare solutions are supporting market expansion worldwide.

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Automated Blood Processing Equipment Industry Demand

[Automated blood processing equipment](#) includes systems and devices used for blood collection, separation, testing, storage, and transfusion management. These technologies are widely used in hospitals, blood banks, pathology laboratories, and diagnostic centers to improve efficiency and reduce manual errors.

Demand for automated blood processing systems is increasing because healthcare facilities are focusing on faster and safer blood management procedures. Rising cases of chronic diseases, cancer, trauma injuries, and blood disorders are increasing the need for blood transfusions and component separation technologies. Automated systems also improve workflow accuracy, reduce contamination risks, and support better inventory management.

The market is further driven by the growing preference for technologically advanced healthcare infrastructure. Automated systems offer long-term cost efficiency by minimizing labor dependency and improving processing speed. In addition, strict government regulations regarding blood safety and quality standards are encouraging healthcare providers to adopt automated equipment.

Automated Blood Processing Equipment Market: Growth Drivers & Key Restraint

Growth Drivers

- **Rising Prevalence of Chronic Diseases:** The increasing number of patients suffering from cancer, cardiovascular diseases, kidney disorders, and anemia has increased the demand for blood transfusions and plasma therapies. This is creating strong demand for automated blood processing technologies in healthcare institutions.
- **Technological Advancements:** Continuous innovations in centrifuges, plasma separators, and automated blood collection systems are improving operational efficiency. Integration of digital monitoring systems and advanced software platforms is further enhancing blood processing accuracy and safety.

- **Increasing Focus on Blood Safety:** Hospitals and blood banks are adopting automated systems to reduce contamination risks and ensure compliance with healthcare standards. Automation also helps improve workflow efficiency and minimizes human errors during blood processing procedures.

Restraint

- **High Equipment Costs:** Advanced automated blood processing systems require significant investment and maintenance expenses. Smaller healthcare facilities in developing regions may face challenges in adopting these technologies due to limited budgets.

Automated Blood Processing Equipment Market: Segment Analysis

Segment Analysis by Product

Needles and syringes remain essential for blood collection and transfusion procedures due to increasing diagnostic testing activities. Blood collection tubes hold a major market share because of their wide application in laboratory testing and automated diagnostic systems. Blood bags are witnessing strong demand due to growing blood storage and transfusion requirements. Lancets are increasingly used in point-of-care testing and diabetes monitoring applications. Filters are gaining importance for blood purification and contamination reduction processes.

Segment Analysis by Type

Plastic blood collection tubes dominate the market because they are lightweight, durable, and compatible with automated laboratory systems. Glass blood collection tubes are mainly used in specialized testing applications where chemical stability is important. Stainless steel blood collection tubes are utilized in advanced medical and research environments requiring high durability. Ceramic blood collection tubes are emerging in niche applications due to their specialized resistance properties.

Segment Analysis by Device Type

Plasma freezers are widely used for preserving plasma quality and supporting long-term storage applications. Hematocrit centrifuges are important for blood component separation and diagnostic testing procedures. Blood warmers are increasingly utilized in surgeries and emergency care to maintain safe blood temperatures during transfusions.

Segment Analysis by System

Automatic blood separators are experiencing strong demand because they efficiently separate blood components with minimal manual involvement. Automatic blood processing systems integrate multiple blood management functions into a single platform, improving productivity and operational safety in hospitals and laboratories.

Segment Analysis by Application

In diagnostics, automated blood processing equipment supports accurate and rapid testing procedures. In treatment applications, these systems are widely used during surgeries, trauma care, and transfusion therapies. Therapeutic applications such as plasma therapy and apheresis procedures are also contributing to market growth.

Segment Analysis by End User

Hospitals represent a major end-user segment due to rising patient admissions and surgical procedures. Blood banks rely on automated systems for safe blood collection and storage management. Pathology laboratories are adopting automation to improve testing efficiency and sample handling. Ambulatory surgical centers are using compact blood processing systems for outpatient procedures, while diagnostic centers are implementing advanced technologies to manage growing testing volumes.

Automated Blood Processing Equipment Market: Regional Insights

North America

North America leads the market due to advanced healthcare infrastructure, high adoption of automation technologies, and strong blood donation networks. Rising chronic disease prevalence and growing investment in diagnostic technologies are supporting market demand in the region.

Europe

Europe holds a significant market share because of strict blood safety regulations and well-established healthcare systems. Increasing aging populations and growing awareness regarding advanced diagnostic solutions are driving regional growth.

Asia-Pacific

Asia-Pacific is expected to witness rapid growth due to improving healthcare infrastructure, increasing healthcare spending, and rising awareness regarding blood safety. Expanding hospital networks and government initiatives supporting diagnostic modernization are also contributing to market development.

Top Players in the Automated Blood Processing Equipment Market

Leading companies operating in the automated blood processing equipment market include Nipro Medical Corporation, BD, Terumo BCT Inc., Fresenius KABI AG, Grifols S.A., Greiner AG, Quest Diagnostics Incorporated, Macopharma, SARSTEDT AG & Co. KG, and ICU Medical, Inc.. These companies are focusing on automation technologies, product innovation, and strategic partnerships to strengthen their market position.

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